

Problem 3

Question

You measure an electric field of 1240 N/C at a point 17.2 cm from a small charged particle.

(a) Calculate the electric flux through a sphere centered on the particle with a radius equal to this distance;

(b) Find the magnitude of the charge.

Enter your answer to Part (a), in N-m²/C, in the box below.

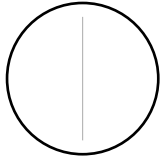
Work

First we will define all the variables we have.

$$E = 1240 \frac{N}{C}$$

$$r = 17.2 \times 10^{-2} m$$

Picture



Part a)

For this, we know the flux through an enclosed surface is equal to the total enclosed charge. This is Gauss's Law, or divergence theorem.